1. Grassmann variables, fermionic coherent states, Gaussian integrals.

- $c|0\rangle = 0$, $c|1\rangle = |0\rangle$, $c^\dagger|0\rangle = |1\rangle$, $c^\dagger|1\rangle = 0$

  $c^\dagger c\,|n\rangle = n|n\rangle$, $\{c, c^\dagger\} = \mathbb{1}$, $c^2 = (c^\dagger)^2 = 0$

$$\psi\psi' + \psi'\psi = 0 \quad = \quad \psi^2 = 0$$

$(n=1)$ $f(\psi) = f_0 + f_1\psi$

$$\frac{\partial f}{\partial \psi} = f_1$$

$(n=2)$ $f(\psi, \psi') = f_{00} + f_{10}\psi + f_{01}\psi' + f_{11}\psi\psi'$

$$\frac{\partial f}{\partial \psi'} = f_{01} - f_{11}\psi \quad , \quad \frac{\partial^2 f}{\partial\psi\partial\psi'} = -f_{11}$$

$$\frac{\partial^2 f}{\partial\psi\partial\psi} = f_{11} = -\frac{\partial^2 f}{\partial\psi\partial\psi'} \quad \Longrightarrow \left\{\frac{\partial}{\partial\psi'}, \frac{\partial}{\partial\psi}\right\} = 0$$

Integration: <u>def</u> $\int d\psi\, 1 = 0$ $\int d\psi\, \psi = 1$

$$\{\psi, c\} = \{\psi, c^\dagger\} = 0$$

- A fermionic coherent state is defined as $c|$ $\rangle =$ $|$ $\rangle$

  $c(x|0\rangle + y|1\rangle) = y|0\rangle \neq \lambda(x|0\rangle + y|1\rangle)$ ∴

  In fact, $|\psi\rangle = |0\rangle - \psi|1\rangle = |0\rangle - \psi c^\dagger|0\rangle = (1 - \psi c^\dagger)|0\rangle$

  $$= e^{-\psi c^\dagger}|0\rangle$$

  这家伙展开后高次项全是0.

$$\begin{cases} c|\psi\rangle = 0 - c\psi|1\rangle = \psi c|1\rangle = \psi|0\rangle \\ \psi|\psi\rangle = \psi|0\rangle - \psi^2|1\rangle = \psi|0\rangle \end{cases} \Longrightarrow c|\psi\rangle = \psi|\psi\rangle$$

similarly, $\langle\psi|$ is a left eigenstate of $c^\dagger$ with $\bar\psi$

$$\langle\psi| = \langle 0|e^{-c\bar\psi} = \langle 0|(1 - c\bar\psi) = \langle 0| - \langle 0|c\bar\psi = \langle 0| - \langle 1|\bar\psi$$

$$\langle\psi|c^\dagger = \langle 0|c^\dagger - \langle 1|\bar\psi c^\dagger = \langle 0|\bar\psi = \langle\psi|\bar\psi$$

ψ 和 $\bar\psi$ 是完全无关的两个东西.

$$\langle\psi|\psi'\rangle = \left(\langle 0| - \langle 1|\bar{\psi}\right)\left(|0\rangle - \psi'|1\rangle\right)$$

$$= 1 + \bar{\psi}\psi' = e^{\bar{\psi}\psi'}$$

↳ 展开以后确实只有两项 ☺

- resolution of unity : $\mathbb{1} = \int d\bar{\psi} \int d\psi\, e^{-\bar{\psi}\psi} |\psi\rangle\langle\bar{\psi}|$

$$\text{LHS} = \int d\bar{\psi}\, d\psi\, (1 - \bar{\psi}\psi)\left(|0\rangle - \psi|1\rangle\right)\left(\langle 0| - \langle 1|\bar{\psi}\right)$$

$$= \int d\bar{\psi}\, d\psi\, (1 - \bar{\psi}\psi)\left(|0\rangle\langle 0| - \psi|1\rangle\langle 0| - \bar{\psi}|0\rangle\langle 1| + \psi\bar{\psi}|1\rangle\langle 1|\right)$$

$$= \int d\bar{\psi}\, d\psi\left(|0\rangle\langle 0| - \psi|1\rangle\langle 0| - \bar{\psi}|0\rangle\langle 1| + \psi\bar{\psi}|1\rangle\langle 1| - \bar{\psi}\psi|0\rangle\langle 0|\right)$$

$$= \int d\bar{\psi}\, d\psi\left(-\bar{\psi}\psi\left(|0\rangle\langle 0| + |1\rangle\langle 1|\right)\right)$$

$$= \int d\bar{\psi}\, d\psi\left(-\bar{\psi}\psi\, \mathbb{1}\right)$$

$$= \int d\bar{\psi}\, \bar{\psi} \int d\psi\, \psi = \mathbb{1}$$

- normally ordered operator $\mathcal{O}(c^\dagger, c)$

$$\langle\psi|H(c^\dagger, c)|\psi'\rangle = H(\bar{\psi}, \psi')\langle\psi|\psi'\rangle = H(\bar{\psi}, \psi')\, e^{\bar{\psi}\psi'}$$

$$\mathrm{Tr}\{\mathcal{O}\} = \sum_{n=0,1} \langle n|\mathcal{O}|n\rangle = \sum_{n=0,1} \iint d\bar{\psi}\, d\psi\, e^{-\bar{\psi}\psi} \langle n|\psi\rangle\langle\psi|\mathcal{O}|n\rangle$$

$$= \iint d\bar{\psi}\, d\psi\, e^{-\bar{\psi}\psi} \sum_{n=0,1} \langle\psi|\mathcal{O}|n\rangle\langle n|-\psi\rangle$$

$|-\psi\rangle = |0\rangle + \psi|1\rangle$

$$= \iint d\bar{\psi}\, d\psi\, e^{-\bar{\psi}\psi} \langle\psi|\mathcal{O}|-\psi\rangle$$

→ 交换了两个 coherent state

- Gaussian integrals over two sets of independent Grassmann variables $\bar{\psi}_j$, ψ_j $j=1,\cdots,N$

$$Z[\bar{\chi},\chi] \equiv \int \prod_{j=1}^{N} [d\bar{\psi}_j\, d\psi_j]\, e^{-\sum_{i,j}^{N} \bar{\psi}_i A_{ij} \psi_j + \sum_j^N \bar{\psi}_j \chi_j + \bar{\chi}_j \psi_j}$$

$$= \det A \cdot e^{\sum_{i,j}^{N} \bar{\chi}_i (A^{-1})_{ij} \chi_j}$$

A: invertible complex matrix, $\bar{\chi}, \chi$: mutually independent (and independent from $\bar{\psi}_j, \psi_j$) sets of Grassmann numbers.

$$* \int \prod_{j=1}^{N} [d\bar{\psi}_j\, d\psi_j]\, e^{-\bar{\psi} A \psi} = \det A$$ 和 Boson 不同

- Wick theorem

Quite similar to bosonic case, with the exception that every combination is multiplied by the parity of the corresponding permutation.

e.g. $$\langle \psi_a \bar{\psi}_b \rangle = \frac{1}{Z[0,0]} \frac{\partial^2 Z[\bar{\chi},\chi]}{\partial \chi_b\, \partial \bar{\chi}_a}\Big|_{\chi=0} = A^{-1}_{ab}$$

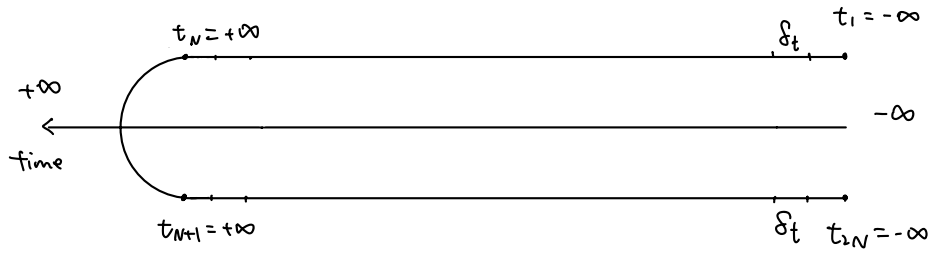
$$\langle \psi_a \psi_b \bar{\psi}_c \bar{\psi}_d \rangle = \frac{1}{Z[0,0]} \frac{\partial^4 Z[\bar{\chi},\chi]}{\partial \chi_d \partial \chi_c \partial \bar{\chi}_b \partial \bar{\chi}_a}\Big|_{\chi=0} = -A^{-1}_{ac} A^{-1}_{bd} + A^{-1}_{ad} A^{-1}_{bc}$$

这一项就比Bosons多个负号

2. Partition functions

Consider a single quantum state with energy ϵ_0. $H = \epsilon_0 c^\dagger c$

$$\mathrm{Tr}\{\rho_0\} = 1 + \rho(\epsilon_0)$$



$$Z = \frac{\mathrm{Tr}\{U_c \rho_0\}}{\mathrm{Tr}\{\rho_0\}} = \frac{\mathrm{Tr}\{U(-\infty,+\infty)\, U(+\infty,-\infty)\, \rho_0\}}{\mathrm{Tr}\{\rho_0\}}$$

$$\mathrm{Tr}\{U_c \rho_0\} = \sum_{n=0,1} \langle n| U_{-\delta_t} U_{-\delta_t} \cdots U_{-\delta_t} U_{+\delta_t} U_{+\delta_t} \cdots U_{+\delta_t} \rho_0 |n\rangle$$

$$= \iint \prod_{j=1}^{2N} [d\bar{\psi}_j d\psi_j]\, e^{-\bar{\psi}_j \psi_j} \sum_n \Big[\langle n|\psi_{2N}\rangle \langle \psi_{2N}|U_{-\delta_t}|\psi_{2N-1}\rangle \langle \psi_{2N-1}|U_{-\delta_t}|\psi_{2N-2}\rangle \cdots$$

$$\cdots \langle \psi_{N+2}|U_{-\delta_t}|\psi_{N+1}\rangle \langle \psi_{N+1}|\psi_N\rangle \langle \psi_N|U_{+\delta_t}|\psi_{N-1}\rangle \langle \psi_{N-1}|U_{+\delta_t}|\psi_{N-2}\rangle$$

$$\cdots \langle \psi_2|U_{+\delta_t}|\psi_1\rangle \langle \psi_1|\rho_0|n\rangle \Big]$$

$$= \iint \prod_{j=1}^{2N} [d\bar{\psi}_j d\psi_j]\, e^{-\bar{\psi}_j \psi_j} \Big[\langle \psi_{2N}|U_{-\delta_t}|\psi_{2N-1}\rangle \cdots \langle \psi_{N+2}|U_{-\delta_t}|\psi_{N+1}\rangle \langle \psi_{N+1}|\psi_N\rangle$$

$$\langle \psi_N|U_{+\delta_t}|\psi_{N-1}\rangle \cdots \langle \psi_2|U_{+\delta_t}|\psi_1\rangle \langle \psi_1|\rho_0|-\psi_{2N}\rangle \Big]$$

↳ GV 交换出的负号

$$* \quad \langle \psi_k|U_{\pm\delta_t}|\psi_{k-1}\rangle = \langle \psi_k| e^{\mp iH(c^\dagger, c)\delta_t} |\psi_{k-1}\rangle$$

$$\approx \langle \psi_k| \, 1 \mp iH(\bar{\psi}_k, \psi_{k-1})\delta_t \, |\psi_{k-1}\rangle$$

$$= e^{\bar{\psi}_k \psi_{k-1}} e^{\mp iH(\bar{\psi}_k, \psi_{k-1})\delta_t} = e^{\bar{\psi}_k \psi_{k-1}} e^{\mp i\epsilon_0 \bar{\psi}_k \psi_{k-1} \delta_t}$$

$$= e^{\bar{\psi}_k \psi_{k-1}} (1 \mp i\epsilon_0 \delta_t)$$

$$* \quad \langle \psi_1|\rho_0|-\psi_{2N}\rangle = -\rho(\epsilon_0)$$

$$\Longrightarrow Z = \frac{1}{\mathrm{Tr}\{\rho_0\}} \iint \prod_{j=1}^{2N} [d\bar{\psi}_j d\psi_j] \exp\Big[i \sum_{j,j'=1}^{2N} \bar{\psi}_j G^{-1}_{jj'} \psi_{j'} \Big]$$

For $N=3$, we have

$$i G^{-1}_{jj'} = \left(\begin{array}{cccc|cccc} -1 & & & & & & & -\rho(\epsilon_0) \\ h_- & -1 & & & & & & \\ & h_- & -1 & & & & & \\ & & 1 & -1 & & & & \\ \hline & & & h_+ & -1 & & & \\ & & & & h_+ & -1 & & \\ & & & & & h_+ & -1 & \end{array}\right)$$

where $h_\pm = 1 \pm i\epsilon_0 \delta_t$

$$\det(iG^{-1}) = 1 + \rho(\epsilon_0)(h_- h_+)^{N-1} = 1 + \rho(\epsilon_0)(1+\epsilon_0^2\delta_t^2)^{N-1}$$

$$\approx 1 + \rho(\epsilon_0) e^{\epsilon_0^2 \delta_t^2 (N-1)} \xrightarrow{N\to\infty} 1 + \rho(\epsilon_0)$$

$$\Longrightarrow Z = \frac{\det(iG^{-1})}{\mathrm{Tr}\{\rho_0\}} = 1 \quad \text{as it should be.}$$

Taking the limit $N\to\infty$ and introduce the continuum notation $\psi_j \to \psi(t)$, we obtain

$$Z = \int D[\bar{\psi}\psi] \exp(iS[\bar{\psi},\psi])$$

$$= \int D[\bar{\psi}\psi] \exp\Big(i\int_C dt\, [\bar{\psi}(t) G^{-1} \psi(t)] \Big)$$

$$\Longrightarrow S[\bar{\psi},\psi] = \sum_{j=1}^{2N} i\bar{\psi}_j\psi_j + \frac{1}{i}(1-i\epsilon_0\delta_t)(\bar{\psi}_2\psi_1 + \cdots + \bar{\psi}_N\psi_{N-1})$$

$$+ \bar{\psi}_{N+1}\psi_N + \frac{1}{i}(1+i\epsilon_0\delta_t)(\bar{\psi}_{N+2}\psi_{N+1} + \cdots + \bar{\psi}_{2N}\psi_{2N-1})$$

$$+ i\bar{\psi}_1\psi_{2N}\rho(\omega_0)$$

$$= \sum_{j=2}^{2N} \Big[i\bar{\psi}_j \frac{\psi_j - \psi_{j-1}}{\delta t_j} - \epsilon_0 \bar{\psi}_j \psi_{j-1} \Big] \delta t_j + i\bar{\psi}_1 [\psi_1 + \rho(\epsilon_0)\psi_{2N}]$$

where $\delta t_j \equiv t_j - t_{j-1} = \pm\delta_t$ 正负号取决于在哪个时间分支上

$$\Longrightarrow G^{-1} = i\partial_t - \epsilon_0$$

$$S[\bar{\psi},\psi]=\int_{-\infty}^{+\infty}dt\left[\bar{\psi}^{+}(t)(i\partial_t-\omega_0)\psi^{+}(t)-\bar{\psi}^{-}(t)(i\partial_t-\omega_0)\psi^{-}(t)\right]$$

这仅是把正负时间分支分开写. 看起来 ψ^+, ψ^- 无关, 其实有关. 因为离散情形下矩阵的 off-diagonal 有非零项.

9.3 Green function and Keldysh rotation

Inverting the discrete matrix, going to the continuum limit, we get

$$\langle \psi^+(t)\, \bar{\psi}^-(t') \rangle \equiv i\, G^<(t,t') = -n_F\, e^{-i\epsilon_0(t-t')} \qquad n_F = \frac{\rho(\epsilon_0)}{1+\rho(\epsilon_0)}$$

$$\langle \psi^-(t)\, \bar{\psi}^+(t') \rangle \equiv i\, G^>(t,t') = (1-n_F)\, e^{-i\epsilon_0(t-t')}$$

$$\langle \psi^+(t)\, \bar{\psi}^+(t') \rangle \equiv i\, G^{\mathcal{T}}(t,t') = \theta(t-t')\, i G^>(t,t') + \theta(t'-t)\, i G^<(t,t')$$

$$\langle \psi^-(t)\, \bar{\psi}^-(t') \rangle \equiv i\, G^{\tilde{\mathcal{T}}}(t,t') = \theta(t'-t)\, i G^>(t,t') + \theta(t-t')\, i G^<(t,t')$$

$$* \quad G^{\mathcal{T}}(t,t') + G^{\tilde{\mathcal{T}}}(t,t') - G^>(t,t') - G^<(t,t') = 0$$

Keldysh rotation:

$$\begin{cases} \psi_1(t) = \frac{1}{\sqrt{2}}\left(\psi^+(t) + \psi^-(t)\right) \\ \psi_2(t) = \frac{1}{\sqrt{2}}\left(\psi^+(t) - \psi^-(t)\right) \end{cases}$$

$$\begin{cases} \bar{\psi}_1(t) = \frac{1}{\sqrt{2}}\left(\bar{\psi}^+(t) - \bar{\psi}^-(t)\right) \\ \bar{\psi}_2(t) = \frac{1}{\sqrt{2}}\left(\bar{\psi}^+(t) + \bar{\psi}^-(t)\right) \end{cases}$$

$$-i \langle \psi_a(t)\, \bar{\psi}_b(t') \rangle = G_{ab}(t,t') = \begin{pmatrix} G^R(t,t') & G^K(t,t') \\ 0 & G^A(t,t') \end{pmatrix}$$

$$G^R(t,t') = -i \langle \psi_1(t)\, \bar{\psi}_1(t') \rangle$$

$$= -i \left\langle \frac{1}{\sqrt{2}}\left(\psi^+(t) + \psi^-(t)\right) \frac{1}{\sqrt{2}}\left(\bar{\psi}^+(t) - \bar{\psi}^-(t)\right) \right\rangle$$

$$= -\frac{i}{2}\left[\langle \psi^+(t)\, \bar{\psi}^+(t) \rangle + \langle \psi^-(t)\, \bar{\psi}^+(t) \rangle - \langle \psi^+(t)\, \bar{\psi}^-(t) \rangle - \langle \psi^-(t)\, \bar{\psi}^-(t) \rangle \right]$$

$$= \frac{1}{2}\left(G^{\mathcal{T}} - G^{\tilde{\mathcal{T}}} + G^> - G^< \right) = \theta(t-t')\left(G^> - G^<\right)$$

$$G^A(t,t') = \frac{1}{2}\left(G^{\mathcal{T}} - G^{\tilde{\mathcal{T}}} - G^> + G^< \right) = \theta(t'-t)\left(G^< - G^>\right)$$

$$G^K(t,t') = \frac{1}{2}\left(G^{\mathcal{T}} + G^{\tilde{\mathcal{T}}} + G^> + G^< \right) = G^> + G^<$$

$$G^A = [G^R]^\dagger \,, \qquad G^K = -[G^K]^\dagger$$

$\dagger$: complex conjugate + $t \leftrightarrow t'$

$$G^A(t,t) + G^R(t,t) = 0$$

$$\Longrightarrow \quad G^R(t,t') = -i\theta(t-t')\, e^{-i\epsilon_0(t-t')} \xrightarrow{FT} (\epsilon - \epsilon_0 + i0)^{-1}$$

$$G^A(t,t') = i\theta(t'-t)\, e^{-i\epsilon_0(t-t')} \xrightarrow{FT} (\epsilon - \epsilon_0 - i0)^{-1}$$

$$G^K(t,t') = -i(1-2n_F)\, e^{-i\epsilon_0(t-t')} \xrightarrow{FT} -2\pi i(1-2n_F)\,\delta(\epsilon-\epsilon_0)$$

$$G^K = G^R \circ F - F \circ G^A \,, \qquad F = F^\dagger$$

9.4 Free fermionic fields and their action